Claire Kim

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PhD Candidate in Computer Science and Applied AI/ML Engineer with experience optimizing large-scale LLM training, inference, and evaluation pipelines on GPU infrastructure. Built enterprise AI solutions at Amazon and AMD spanning AI agents, post-training and alignment techniques (RLHF, LoRA, PEFT), RAG, and scalable ML platforms. Experienced partnering with cross-functional engineering, research, and stakeholder teams to translate performance and reliability challenges into production AI solutions.

EDUCATION

PhD, Computer Science **2021 – 2026**

Arizona State University, Tempe, AZ

- Advisors: Dr. Huan Liu and Dr. Mickey Mancenido

M.E., Computer Science & Engineering **2017 – 2019**

B.S., Computer Science & Engineering **2013 – 2017**

Korea University, Seoul, South Korea

WORK EXPERIENCE

Applied AI Engineer — Sep 2026 –

Cognizant — **Remote**

- Client: Verizon AI Platform; contract via Intellan Technologies LLC

Applied Scientist Intern — Sep – Dec 2025

Amazon — **Bellevue, WA**

- Improved production sentiment forecasting models supporting community operations at Amazon by developing advanced sequence modeling approaches and improving prediction robustness at scale.
- Designed and productionized an end-to-end ML pipeline spanning data collection, model training, experiment tracking, and automated deployment, and monitoring using Python, AWS, MLflow, and CI/CD pipelines.

AI/ML Intern — May – Aug 2025

AMD — **Austin, TX**

- Architected and deployed Q-RAG, an enterprise question-centric Retrieval-Augmented Generation framework that leveraged synthetic QA generation and LLM-as-a-Judge evaluation to improve knowledge retrieval, identify documentation gaps, and reduce hallucinations across engineering support systems.
- Designed scalable AI evaluation infrastructure enabling automated benchmarking of enterprise RAG systems while significantly reducing manual expert review.
- Partnered with engineering stakeholders to identify knowledge management challenges, prototype AI solutions, iterate from user feedback, and deliver production-ready LLM workflows.

Software Development Intern — Aug – Dec 2024

AMD — **Austin, TX**

- Designed production AI inference architecture supporting enterprise RAG, multi-agent orchestration, user-feedback integration, and large-scale experimentation.
- Optimized AI inference and data pipelines to improve throughput, reduce evaluation latency, and eliminate performance bottlenecks across scalable production deployments.
- Collaborated with AI engineers and domain experts to rapidly prototype, iterate, and deploy enterprise AI capabilities.

Graduate Research Assistant — May 2022 – Aug 2024

DHS-CAOE (Center for Accelerating Operational Efficiency) — **Tempe, AZ**

- Designed NLP solutions for topic modeling and text summarization using BERT and Llama-2.
- Partnered with researchers, domain experts, and software engineers to architect a trustworthy multi-agent RAG system supporting AI-assisted intelligence analysis.
- Delivered an interactive analytics dashboard enabling stakeholders to explore AI-generated insights through web-based visualizations.

Graduate Research Assistant — Jan 2021 – Aug 2022

ONR (Office of Naval Research Project), Arizona State University — **Tempe, AZ**

- Researched the integration and mutual influence of online and offline COVID-19 datasets using topic modeling.
- Analyzed 2M tweets for sentiment and stance detection in pandemic-related discussions.

SELECTED PROJECTS

Adaptive Triggering for Bias Correction in LLM Reasoning (Under Review)

- Formulated inference-time bias correction as an online change-point detection problem, using a trigger to decide when a developing LLM reasoning trajectory shows enough evidence of stereotype reliance to warrant correction.
- Designed and evaluated white-box (logit-based) and black-box (LLM-judge) bias-risk signals across seven open-weight models and GPT-4o-mini, running large-scale experiments on ASU's Sol supercomputer.
- Recovered most of the accuracy lost under fixed-interval intervention (90.1% vs. 82.9%, baseline 92.1%) while cutting interventions ~40%, validated via independent judge and statistical testing.
- Corrected four undocumented schema errors in the BBQ bias benchmark that had left 17–50% of items in affected categories mis-scored, and open-sourced the fix.

TECHNICAL SKILLS

Large Language Models & Agents: Retrieval-Augmented Generation (RAG), Agentic AI, Multi-Agent Systems, LangChain, LlamaIndex, LLM-as-a-Judge, Prompt Engineering, Synthetic Data Generation, Inference-Time Scaling
Model Training & Alignment: Model Alignment, Post-training, RLHF, RLAIF, LoRA, PEFT, Responsible AI, AI Safety
Retrieval & Search: Vector Search, Knowledge Retrieval, Embeddings
Machine Learning: PyTorch, TensorFlow, Scikit-Learn, Hugging Face, Pandas, NumPy
Cloud & Deployment: AWS, GCP, Docker, MLflow, TensorRT, Model Serving, GPU Optimization, CI/CD
Software Engineering: Flask, Node.js, REST APIs, Git, Linux, Unit Testing
Languages: Python, SQL, Java, JavaScript, Bash

SELECTED PUBLICATIONS | Google Scholar

AI Magazine'25 PADTHAI-MM: A Principled Approach for the Design of Trustworthy, Human-Centered AI systems using the MAST Methodology

ASONAM'24 Robust Stance Detection: Understanding Public Perceptions in Social Media

JAIR'23 Evaluating Trustworthiness of AI-Enabled Decision Support Systems: Validation of the Multisource AI Scorecard Table (MAST)

ACADEMIC SERVICE & TEACHING

Reviewer at **AMLC 2025** (Amazon Gen AI Evaluation Workshop) — 2025
Invited Reviewer for **NeurIPS 2025** (Reliable ML Workshop) — 2025
Program Committee (PC) member of **ASONAM 2024** conference — 2024
Invited talk at Machine Learning Day 2024, Arizona State University — 2024
Invited talk at SCAI AI Day, Arizona State University — 2023

Program Committee (PC) member of **ASONAM**, **SBP-BRiMS 2023** conferences	2023
Invited Reviewer for **EMNLP 2023** conference	2023
Reviewer at **ECML-PKDD, ACM MultiMedia, ASONAM,** and **AAAI** conferences	2022
Graduate Teaching Assistant, Object-Oriented Programming & Data Structures (ASU)	2022